

LARAVEL

You're developing an e-commerce system using Laravel where products can have discounts based on different strategies (e.g., fixed amount, percentage, or loyalty-based). You are required to design a flexible and extensible discount system using OOP principles.

Implement a **DiscountService** that can apply different discount strategies to a product's price.

1. Create an interface named **DiscountStrategyInterface** with a method: (1 marks)

```
public function calculate(float $originalPrice): float;
```

[1 mark]

2. Implement three discount strategy classes from **DiscountStrategyInterface**:

- a. **FixedDiscount** (original price – discount amount)
- b. **PercentageDiscount** (original price – [percentage * original price])
- c. **LoyaltyDiscount** (original price – [0.85 * original price])

[3 marks]

3. Create a service class called **DiscountService** that:

- a. Accepts any **DiscountStrategyInterface** via constructor injection.
- b. Has a method **applyDiscount(float \$price): float** that uses the injected strategy.
- c. Add validation to the input value (throw **HTTP 400 – Bad Request**):
 - i. Percentage discount, must between 0-100.
 - ii. Fixed discount value, must not more than the original value.

[3 marks]

FLUTTER

Design a Flutter project app using OOP principles to display a list of adoptable pets, allow filtering by pet type (e.g., Dog, Cat), and mark pets as adopted.

1. Model Classes (OOP Focus)

a. Create an abstract **Pet** class with common property:

- i. **name**
- ii. **age**
- iii. **breed**
- iv. **isAdopted**

[1 mark]

b. Create **Dog** and **Cat** subclasses derived from **Pet** class, each with at least one unique property.

- i. **Dog** class has **isTrained** property (default is false).
- ii. **Cat** class has **isIndoor** property (default is true).

[2 marks]

2. Service Layer

a. Create a **PetRepository** class that returns a list of mixed Pet objects (example as follow):

```
Dog(name: 'Rex', age: 2, breed: 'Labrador', isTrained: true),  
Dog(name: 'Buddy', age: 1, breed: 'Beagle'),  
Cat(name: 'Mittens', age: 3, breed: 'Siamese', isIndoor: false),  
Cat(name: 'Whiskers', age: 4, breed: 'Persian'),
```

[2 mark]